

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

# Chapter 8: Remainder and Factor Theorems

Topic-wise short notes with solved examples

These two theorems give the remainder of a polynomial division without dividing, and tell whether a linear expression is a factor. They are then used to factorise cubic polynomials.

## 1 Polynomials and division

- A **polynomial** in  $x$  is written  $f(x)$ . Its value at  $x = a$  is  $f(a)$ , found by substituting  $a$  for  $x$ .
- When  $f(x)$  is divided by a divisor, we get a quotient and a remainder:

$$\text{Dividend} = \text{divisor} \times \text{quotient} + \text{remainder}$$

- When the divisor is linear (degree 1), the remainder is a constant.

## 2 The remainder theorem

- **Theorem:** if  $f(x)$  is divided by  $(x - a)$ , the remainder is  $f(a)$ .

| Divisor  | Remainder                    |
|----------|------------------------------|
| $x - a$  | $f(a)$                       |
| $x + a$  | $f(-a)$                      |
| $ax - b$ | $f\left(\frac{b}{a}\right)$  |
| $ax + b$ | $f\left(-\frac{b}{a}\right)$ |

- Rule: put the divisor equal to zero, solve for  $x$ , and substitute that value in  $f(x)$ .

**Example 2.1** Find the remainder when  $f(x) = x^3 - 3x^2 + 4x - 5$  is divided by  $(x - 2)$ .

**SOLUTION**

$$x - 2 = 0 \text{ gives } x = 2$$

$$f(2) = 8 - 12 + 8 - 5$$

$$\text{Remainder} = -1$$

**Example 2.2** Find the remainder when  $f(x) = 2x^3 + 3x^2 - 5x + 1$  is divided by  $(2x + 1)$ .

SOLUTION

$$2x + 1 = 0 \text{ gives } x = -\frac{1}{2}$$

$$f\left(-\frac{1}{2}\right) = 2\left(-\frac{1}{8}\right) + 3\left(\frac{1}{4}\right) + \frac{5}{2} + 1 = -\frac{1}{4} + \frac{3}{4} + \frac{5}{2} + 1$$

**Remainder = 4**

### 3 The factor theorem

- **Theorem:**  $(x - a)$  is a factor of  $f(x)$  if and only if  $f(a) = 0$ .
- So a divisor is a factor exactly when the remainder is zero.
- Similarly  $(ax + b)$  is a factor of  $f(x)$  if  $f\left(-\frac{b}{a}\right) = 0$ .

**Example 3.1** Show that  $(x - 2)$  is a factor of  $x^3 - 7x + 6$ .

SOLUTION

$$f(2) = 8 - 14 + 6 = 0$$

**Since  $f(2) = 0$ ,  $(x - 2)$  is a factor.**

**Example 3.2** Find the value of  $a$  if  $(x - 2)$  is a factor of  $x^3 + ax^2 - 4x + 4$ .

SOLUTION

$$f(2) = 0: 8 + 4a - 8 + 4 = 0$$

$$4a = -4$$

$$a = -1$$

### 4 Finding unknown constants

- Each condition (a factor, or a given remainder) gives one equation. Two unknowns need two conditions.

**Example 4.1** The polynomial  $f(x) = x^3 + ax^2 + bx - 6$  has  $(x - 1)$  as a factor and leaves remainder 12 when divided by  $(x - 3)$ . Find  $a$  and  $b$ .

SOLUTION

$$f(1) = 0: 1 + a + b - 6 = 0, \text{ so } a + b = 5$$

$$f(3) = 12: 27 + 9a + 3b - 6 = 12, \text{ so } 9a + 3b = -9 \text{ and } 3a + b = -3$$

$$\text{Subtract: } 2a = -8, \text{ so } a = -4 \text{ and } b = 9$$

$$a = -4, b = 9$$

## 5 Factorising a cubic polynomial

- **Step 1:** find one factor by trial. Try  $x = 1, -1, 2, -2, 3, -3, \dots$ , which are factors of the constant term, until  $f(a) = 0$ . Then  $(x - a)$  is a factor.
- **Step 2:** divide  $f(x)$  by  $(x - a)$  to get a quadratic quotient.
- **Step 3:** factorise the quadratic.

**Example 5.1** Factorise  $x^3 - 7x + 6$  completely.

SOLUTION

$f(1) = 1 - 7 + 6 = 0$ , so  $(x - 1)$  is a factor.

Dividing:  $x^3 - 7x + 6 = (x - 1)(x^2 + x - 6)$

$x^2 + x - 6 = (x + 3)(x - 2)$

$x^3 - 7x + 6 = (x - 1)(x - 2)(x + 3)$

**Example 5.2** Factorise  $2x^3 + x^2 - 13x + 6$  completely, and hence solve  $2x^3 + x^2 - 13x + 6 = 0$ .

SOLUTION

$f(2) = 16 + 4 - 26 + 6 = 0$ , so  $(x - 2)$  is a factor.

Dividing:  $2x^3 + x^2 - 13x + 6 = (x - 2)(2x^2 + 5x - 3)$

$2x^2 + 5x - 3 = (2x - 1)(x + 3)$

$(x - 2)(2x - 1)(x + 3)$ ; the solutions are  $x = 2$ ,  $x = \frac{1}{2}$  and  $x = -3$ .

### REMEMBER

Quick tests: if the sum of all the coefficients is 0, then  $(x - 1)$  is a factor. If the sum of the coefficients of the even powers equals the sum of the coefficients of the odd powers, then  $(x + 1)$  is a factor.

## 6 Common mistakes

- Substituting  $x = a$  for the divisor  $(x + a)$ . The value to use is  $x = -a$ .
- Sign errors with odd powers of a negative number:  $(-2)^3 = -8$ .
- Stopping after one factor. "Factorise completely" needs all three linear factors.
- Leaving out a missing term in long division. Write  $x^3 + 0x^2 - 7x + 6$ .